

EXERCISE 6

COMPUTATIONAL NUMBER THEORY AND CRYPTOGRAPHIC APPLICATIONS

1. Using the basic form of Euclid's algorithm, compute the greatest common divisor of

(1) 7469 and 2464

(2) 2689 and 4001

For this problem use only a pocket calculator. Show every iteration step of Euclid's algorithm, i.e., don't write just the answer, which is only a number. Also, for every gcd, provide the chain of gcd computations, i.e.,

$$\gcd(r_0, r_1) = \gcd(r_1, r_2) = \dots$$

2. Using the extended Euclidean algorithm, compute the greatest common divisor and the parameters s, t of

(1) 198 and 243

(2) 1819 and 3587

For every problem check if $sr_0 + tr_1 = \gcd(r_0, r_1)$ is actually fulfilled. The rules are the same as above: use a pocket calculator and show what happens in every iteration step.

3. With the Euclidean algorithm we finally have an efficient algorithm for finding the multiplicative inverse in $\mathbb{Z}_m$ that is much better than exhaustive search. Find the inverses in $\mathbb{Z}_m$ of the following elements a modulo m :

(1) $a = 7, m = 26$ (affine cipher)

(2) $a = 19, m = 999$

Note that the inverses must again be elements in $\mathbb{Z}_m$ and that you can easily verify your answers.

4. Determine $\varphi(m)$, for $m = 12, 15, 26$, according to the definition: Check for each positive integer n smaller m whether $\gcd(n, m) = 1$. (You do not have to apply Euclid's algorithm.)

5. Develop formulae for $\varphi(m)$ for the special cases when

(1) m is a prime

(2) $m = p \cdot q$, where p and q are primes. This case is of great importance for the RSA cryptosystem. Verify your formula for $m = 15, 26$ with the results from the previous problem.

6. Compute the inverse $a^{-1} \bmod n$ with Fermat's Theorem (if applicable) or Euler's Theorem:

$$a = 4, n = 7$$

$$a = 5, n = 12$$

$$a = 6, n = 13$$

7. Verify that Euler's Theorem holds in $\mathbb{Z}_m$, $m = 6, 9$, for all elements a for which $\gcd(a, m) = 1$. Also verify that the theorem does not hold for elements a for which $\gcd(a, m) \neq 1$.

8. For the affine cipher in Chapter 1 the multiplicative inverse of an element modulo 26 can be found as

$$a^{-1} \equiv a^{11} \bmod 26.$$

Derive this relationship by using Euler's Theorem.

9. The extended Euclidean algorithm has the initial conditions $s_0 = 1, s_1 = 0, t_0 = 0, t_1 = 1$. Derive these conditions. It is helpful to look at how the general iteration formula for the Euclidean algorithm was derived in this chapter.